

Osmosis Practice Problems

*Good practice for test-taking
strategy, too.*

#1

– If you soak your hands in dishwater, you may notice that your skin absorbs water and swells into wrinkles. This is because your skin cells are _____ to the _____ dishwater.

- A. Hypotonic...hypertonic
- B. Hypertonic...hypotonic
- C. Hypotonic...hypotonic
- D. Isotonic...hypotonic
- E. Hypertonic...isotonic

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- B. **Hypertonic...hypotonic**
- C. Hypotonic...hypotonic
- D. Isotonic...hypotonic
- E. Hypertonic...isotonic

#2

- You decide to buy a new fish for your freshwater aquarium. When you introduce the fish into its new tank, the fish swells up and dies. You later learn that it was a fish from the ocean.
- Based on what you know of tonicity, the most likely explanation is that the unfortunate fish went from a(n) _____ solution into a(n) _____ solution.
 - isotonic, hypotonic
 - hypertonic, isotonic
 - hypotonic, hypertonic
 - hypotonic, isotonic
 - isotonic, hypertonic

#2

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 - B. hypertonic, isotonic
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 - D. hypotonic, isotonic
 - E. isotonic, hypertonic

#3

- In osmosis, water always moves toward the _____ solution: that is, toward the solution with the _____ solute concentration.
 - A. isotonic, greater
 - B. hypertonic, greater
 - C. hypertonic, lesser
 - D. hypotonic, greater
 - E. hypotonic, lesser

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#4

- The concentration of solutes in a red blood cell is about 2%. Sucrose cannot pass through the membrane, but water and urea can. Osmosis would cause red blood cells to shrink the most when immersed in which of the following solutions?
 - A. a hypertonic sucrose solution
 - B. a hypotonic sucrose solution
 - C. a hypertonic urea solution
 - D. a hypotonic urea solution
 - E. pure water

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 - D. a hypotonic urea solution
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#5

- Sea water is dangerous to drink because
 - A. one cup of sea water contains enough sodium to poison you.
 - B. sea water is hypertonic to your body tissues and drinking it will cause you to lose water.
 - C. sea water is isotonic to your body fluids and you will absorb too much water.
 - D. the salt causes hypertension and you will promptly die of a stroke.
 - E. it contains toxic levels of iodine.

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#6

- If the volume of a cell increases when it is placed in a solution, that solution is said to be _____ to the cell.

- A. hypertonic
- B. subatomic
- C. isotonic
- D. gin and tonic
- E. hypotonic

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#7

- Inside one osmosis bag*, A, is a 50% glucose solution and inside bag B is a 20% glucose solution. Both bags are put into beakers containing 100% water.
 - *Osmosis bags are membranes that let water through but not glucose.
- A. Bag A will gain weight.
- B. Bag B will gain weight.
- C. Both bags will gain weight.
- D. Both bags will lose weight.
- E. Both bags will remain the same.

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#8

- A 0.9% NaCl solution is isotonic to red blood cells. Which of these describes the results if red blood cells are placed into a 9% solution of NaCl?
 - A. They will burst.
 - B. They will shrink.
 - C. Nothing will happen.
 - D. They will expand but not burst.
 - E. None of the above.

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 - C. Nothing will happen.
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 - E. None of the above.

#9

- Wallway is a new general herbicide for aquatic plants. Its main ingredient is a marine salt solution. It is effective against freshwater but not saltwater plants. It works by breaking down the cell walls of the plants. The freshwater plants die because their cells
 - A. swell and cease to function.
 - B. shrink.
 - C. remain the same size but malfunction.
 - D. are crushed by the weight of the plant.

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#10

- A red blood cell has a salt concentration of 0.9%. What will happen if it is placed into a 1% salt solution? The red blood cell will
 - A. shrink if its membrane is permeable to both the salt and the water.
 - B. shrink if its membrane is impermeable to the salt and permeable to the water.
 - C. maintain its shape - nothing will happen.
 - D. swell and probably burst because its membrane is impermeable to salt and permeable to water.
 - E. swell and probably burst because its membrane is impermeable to water and permeable to salt.

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#11

- Flasks X, Y, and Z contain solutions with different concentrations of the solute NaCl.
 - Flask X has 0.5% NaCl
 - Flask Y has 0.9% NaCl
 - Flask Z has 1.5% NaCl
- Red blood cells (0.9% NaCl) will be placed into each flask. Predict what will happen to the blood cells in each of the flasks.

A.	Flask X: Contract	Flask Y: Unchanged	Flask Z: Swell
B.	Flask X: Swell	Flask Y: Unchanged	Flask Z: Contract
C.	Flask X: Unchanged	Flask Y: Swell	Flask Z: Contract
D.	Flask X: Unchanged	Flask Y: Unchanged	Flask Z: Unchanged

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C.	Flask X: Unchanged	Flask Y: Swell	Flask Z: Contract
D.	Flask X: Unchanged Unchanged	Flask Y: Unchanged	Flask Z: